RIEMANNIAN GEOMETRY (MATH 319)

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1. Introduction

A *Riemannian Manifold* is a manifold M that is also a metric space with some metric g (with some additional properties). We will see that such a manifold has a "curvature" $K(g)$. The basic questions in this subject are:

(1) What does $K(g)$ tell us about M?
(2) Given M and varying g, what can $K(g)$ be?

An example of the first type of question comes from physics: the Einstein Equation.

The second question is solved in dimensions 2 and 3, which are the results of Riemann Uniformization and the Poincaré Geometrization Conjecture respectively.

1.1. Riemannian Metrics.

More precisely, a Riemannian Manifold must have the following two properties of its metric g:

(i) g comes from a dot product on TM, i.e. we have bilinear maps $g_p : T_pM \times T_pM \to \mathbb{R}$ at each $p \in M$.
(ii) For any chart $\varphi : M \to \mathbb{R}^n$, $g_x(\partial_i\varphi, \partial_j\varphi)$ is smooth in x.

Given such a g, one can define the *length* of curves on M by integrating the length-form over the curve, and similarly we can define the volume of a subset of M.

The important equivalence class for such manifolds is *isometry*. A *local isometry* is a map $F : (M, g) \to (N, h)$ such that $F^*h = g$. It is a *global isometry* if F is a diffeomorphism. "Almost all" metrics have no nontrivial isometries, but they are common in examples we care about.

Examples:

- *Flat*: $M = \mathbb{R}^n$, standard inner product.
- *Spherical*: $M = S^n$, standard inner product induced by inclusion in $\mathbb{R}^{n+1}$.
- *Hyperbolic*: $M = B^n$,
$$h_p(\vec{x}, \vec{y}) = \frac{4}{(1 - |p|^2)^2}\langle \vec{x}, \vec{y}\rangle.$$

The isometries of $\mathbb{R}^n$ and S^n are $\mathrm{O}_n(\mathbb{R})$ + affine transformations and $\mathrm{O}_n(\mathbb{R})$ respectively. For hyperbolic space, the isometries are conformal maps.

If G is a discrete subgroup of the group of isometries of (M, g) which acts on M freely and properly discontinuously, then we have a *quotient manifold*
$$M/G := \frac{M}{\{x \sim \varphi(x) \;\; \varphi \in G\}}.$$

Examples of Quotients:

- Take some $\mathbb{Z}^n$ as a subgroup of affine transformations on $\mathbb{R}^n$, for which $\mathbb{R}^n/\mathbb{Z}^n = T^n$, the n-torus.
- Let $M = S^3$ treated as a subspace of $\mathbb{C}^2$. Let φ be a rotation in each complex component with periods p, q that are *coprime*, and we have
$$L_{p,q} := S^3/\langle\varphi\rangle$$
called the "Lens Space" or sometimes "Poincaré Sphere."

1.2. Connections.

Let $\mathfrak{X}(M)$ be the space of smooth vector fields on M. $\mathfrak{X}(M)$ has the structure of a vector space with coefficients in $C^\infty(M)$. Moreover, there is also an action of $\mathfrak{X}(M)$ on $C^\infty(M)$, given by $Vf := \partial_V f$. The Riemannian metric on M gives us an inner product $\langle\bullet, \bullet\rangle : \mathfrak{X}(M)^2 \to C^\infty(M)$.

A *connection* is a map $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ satisfying

- Linearity in the first argument: $\nabla_{f\cdot X}Y = f \cdot \nabla_X Y$.
- $\nabla_X(f \cdot Y) = \partial_X f \cdot Y + f \cdot \nabla_X Y$ (Leibniz rule)

One should think of $\nabla_X Y$ as the "derivative of Y as it flows along X." Connections allow us to assign a derivative for vector fields along curves: if V is a vector field defined on the curve $c : I \to M$, the "covariant derivative" is another vector field on c denoted $DV/\mathrm{d}t$ and defined

$$\frac{DV}{\mathrm{d}t}(t_0) = (\nabla_{(\mathrm{d}c/\mathrm{d}t)^{ext}} V^{ext})(c(t_0))$$

where $(\mathrm{d}c/\mathrm{d}t)^{ext}$ and V^{ext} are extensions of $\mathrm{d}c/\mathrm{d}t, V$ to a neighborhood of $c(t_0)$.

Lemma (Properties of covariant derivative): $DV/\mathrm{d}t$ is the unique vector field satisfying the following:

(a) Linearity in V: $D/\mathrm{d}t(V + W) = DV/\mathrm{d}t + DW/\mathrm{d}t$.
(b) Leibniz rule: $D/\mathrm{d}t(f \cdot V) = \mathrm{d}f/\mathrm{d}t \cdot V + f \cdot DV/\mathrm{d}t$.
(c) If V is defined on all of M, then $DV/\mathrm{d}t = \nabla_{\mathrm{d}c/\mathrm{d}t} V$.

Proof. do later □

Given a connection ∇, one can define the *torsion* and *curvature*

- $T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y]$
- $R(X, Y)Z := (\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]})(Z)$

both of which are $C^\infty(M)$-linear in all arguments (unlike $\nabla_X Y$ itself). For example,

$$\begin{aligned} T(fX, Y) &= \nabla_{fX} Y - \nabla_Y (fX) - [fX, Y] \\ &= f \cdot \nabla_X Y - (\partial_Y f \cdot X + f \cdot \nabla_Y X) - (\partial_Y f \cdot X + f \cdot [X, Y]) \\ &= f \cdot \nabla_X Y - f \cdot \nabla_Y X - f \cdot [X, Y] \\ &= f \cdot T(X, Y). \end{aligned}$$

In general there are many connections ∇ on any given Riemannian manifold, so it would be nice to have some canonical choice. And there is one ∇ that is especially good:

Theorem (Levi-Civita): Given (M, g), there is a unique connection ∇ which is

- *Symmetric*, meaning $\nabla_X Y - \nabla_Y X = [X, Y]$ (equivalently $T(X, Y) = 0$ for all X, Y).
- *Compatible* with g, meaning $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$ (the product rule).

Such a ∇ is called a *Levi-Civita* or *Riemannian* connection.

Proof. We can show with some algebra that the two conditions are equivalent to

$$\langle Z, \nabla_Y X\rangle = \frac{1}{2}\Big(X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle [X, Z], Y\rangle - \langle [Y, Z], X\rangle - \langle [X, Y], Z\rangle\Big)$$

so the unique connection is both given and determined by this equation, called the *Koszul formula.* □

From now on we'll assume that ∇ is the Levi-Civita connection.

To actually compute $\nabla_A B$ at $p \in M$ for two given vector fields $A, B \in \mathfrak{X}(M)$, we may want to choose local coordinates in a neighborhood U of p. If the local coordinates are $x^1, \ldots, x^n$ and the vector fields at p are given the basis $\partial_1, \ldots, \partial_n$, then we can compute $(\nabla_A B)_p$ by writing $A = \sum_i a_i \partial_i$ and $B = \sum_i b_i \partial_i$ $(a_i, b_i \in C^\infty(U))$ and using the properties of ∇:

$$\begin{aligned} \nabla_A B &= \sum_i a_i \nabla_{\partial_i} B \\ &= \sum_i a_i \sum_j b_j \nabla_{\partial_i} \partial_j + \partial_i b_j \cdot \partial_j \\ &= \Big(\sum_{ij} a_i b_j \nabla_{\partial_i} \partial_j\Big) + \Big(\sum_{ij} a_i \partial_i b_j \cdot \partial_j\Big). \end{aligned}$$

These functions $\nabla_{\partial_i}\partial_j$ are important, so we abbreviate

$$\nabla_{\partial_i}\partial_j =: \sum_k \Gamma_{ij}^k \partial_k.$$

The coefficient function $\Gamma_{ij}^k \in C^\infty(U)$ is called the *Christoffel Symbol*, and it depends only on g. Note that if ∇ is the Levi-Civita connection, then $\Gamma_{ij}^k = \Gamma_{ji}^k$, as $[\partial_i, \partial_j] = 0$. In terms of Γ_{ij}^k we have

$$\nabla_A B = \sum_k \left(\sum_{ij} a_i b_j \Gamma_{ij}^k + \partial_A b_k \right) \cdot \partial_k.$$

This shows that $(\nabla_A B)_p$ depends only on $A(p), B(p)$, and $\partial_{A(p)} b_k$. In particular, $\nabla_A B$ depends on $A(p)$ but not any other values of A, whereas for B the local behavior near p matters too.

In the specific case of $\mathbb{R}^n$ with the Euclidean metric, $\Gamma_{ij}^k = 0$, so this becomes

$$\nabla_A^{\mathbb{R}^n} B = \sum_k \partial_A b_k \partial_k.$$

Another notable example is on S^n, where the Levi-Civita connection is

$$\nabla_A^{S^n} B = (\nabla_{A^{ext}}^{\mathbb{R}^{n+1}} B^{ext})^T$$

where A^{ext}, B^{ext} are A, B extended to $\mathbb{R}^{n+1}$ and X^T is the orthogonal projection of $X(p)$ onto $T_p S^n$.

To compute the Christoffel symbols themselves, we have (following from the Levi-Civita theorem)

$$\Gamma_{ij}^k = \frac{1}{2} \sum_\ell \Big(\partial_i g_{j\ell} + \partial_j g_{\ell i} - \partial_\ell g_{ij} \Big) g^{\ell k}$$

where $g_{ij} := \langle \partial_i, \partial_j \rangle$, a smooth function on U for each i, j, and g^{ij} is defined as the i, j entry in the inverse matrix $(g^{ij}) = (g_{ij})^{-1}$.

1.3. **Geodesics and the Exponential Map.** In a Riemannian manifold, given a curve $\gamma : I \to M$ we have a γ' and $\gamma'' = \nabla_{\gamma'}\gamma'$ given by the Levi-Civita connection. γ is called a *geodesic* when $\gamma'' = 0$. This implies that γ can be parameterized by arc length, as $|\gamma'|$ is constant.

Another slightly broader definition of geodesic is a curve $\gamma : I \to M$ for which

$$\gamma''(t) = \lambda(t)\gamma'(y)$$

which means, equivalently, that γ can be re-parameterized such that $\gamma''(t) = 0$.

Examples:

- Let $M = S^n$. The geodesics are curves contained in the great circles. For these curves $\gamma'' = -\gamma$. There is an acceleration but it is "invisible to the sphere" because it is orthogonal to the tangent space.
- Let $M = B^n$ with the hyperbolic metric. A straight line away from the origin can be re-parameterized to be a geodesic, by taking the distance to be $\tanh(t)$. Away from the origin, the geodesics are parts of circles meeting the boundary orthogonally.

Proposition: In general if γ is a curve on M and there is an isometry $P : M \to M$ for which $P(\gamma(t)) = \gamma(t)$ but P reverses orientation of normal vectors to γ at each point, then γ is a geodesic.

On hyperbolic space, the isometry corresponding to each geodesic is like an involution.

Theorem (Existence and Uniqueness of Geodesics): For each $p \in M$ and $v \in T_p M$, locally there is a unique curve $\gamma : (-\varepsilon, \varepsilon) \to M$ which is a geodesic, $\gamma(0) = p$, and $\gamma'(0) = v$. We call this $\gamma(t) =: \exp_x(tv)$.

Proof. If $\varphi : U \to \mathbb{R}^n$ is a chart near p with local coordinates $x_1, \ldots, x_n$, then the condition $D\gamma'/\mathrm{d}t = 0$ just becomes a second-order ODE involving Γ_{ij}^k:

$$x_k'' + \sum_{ij} x_i' x_j' \Gamma_{ij}^k = 0 \quad \forall 1 \leq k \leq n$$

with initial conditions on $\gamma(0)$ and $\gamma'(0)$, so there is a unique solution. □

The map $\exp_x$ is very nice. We'll now show some of its properties.

Gauss's Lemma: Fixing $p \in M$, for all $v, w \in T_pM$ with $|v| < \varepsilon$,

$$\langle (\mathrm{d}\exp)_p(v), (\mathrm{d}\exp)_p(w) \rangle = \langle v, w \rangle$$

In other words, $\exp_p$ is locally a conformal map.

Proof. By linearity we can break into cases of orthogonal and parallel. If $w = \lambda v$ then letting $\gamma(t) = \exp(tv)$,

$$\mathrm{d}\exp(\lambda v) = \lambda \gamma'(1),$$

so we have

$$\langle \mathrm{d}\exp(v), \mathrm{d}\exp(w) \rangle = \lambda \langle \gamma'(1), \gamma'(1) \rangle = \lambda |v|^2 = \langle v, w \rangle$$

On the other hand, if $w \perp v$, and wlog $|w| = |v|$, then let $w : (-\varepsilon, \varepsilon) \to T_pM$ be the circular curve with $v(0) = v$, $v'(0) = w$, and $|v(t)| = |v|$ for all t. Now let $F : (\varepsilon, \varepsilon) \times (0, 1) \to M$ be defined

$$F(s, t) := \exp_p(t \cdot v(s)).$$

For each fixed s, $F(s, \bullet)$ is a geodesic curve in M, so $F(s, t)$ is sort of a "geodesic fan" spreading out from p. We want to show $\langle \mathrm{d}\exp(v), \mathrm{d}\exp(w) \rangle = 0$, and F is relevant to this because

$$\partial_t F(0, 1) = \mathrm{d}\exp(v), \quad \partial_s F(0, 1) = \mathrm{d}\exp(w)$$

so we need to show $\langle \partial_t(0, 1), \partial_s(0, 1) \rangle = 0$. This can be done with some slightly onerous computation which I will omit. □

As geodesics sprout from x in all directions, it may be that some of them intersect, and it may be that they don't. As it turns out, they always go a non-zero distance before intersecting.

Proposition (Normal Neighborhood): For $p \in M$, there is a neighborhood U and $\varepsilon < r$ (for some r called the *injectivity radius* at x) for which:

(i) $\exp_q : B_\varepsilon(U) \to M$ is a diffeomorphism onto its image.
(ii) For all $q_1, q_2 \in U$, there is a geodesic connecting q_1 and q_2 (not necessarily contained in U).

Proof. Fix a local chart $\varphi : \tilde{U} \to M$ near p so that $\varphi(0) = p$. By the previous theorem, for some $V - 1 \subset \tilde{U}$ and $\varepsilon_1 > 0$ we have a well-defined map $F : V_1 \times B_{\varepsilon_1} \to M \times M$ where

$$F(x, v) = (\varphi(x), \exp_{\varphi(x)}(v)).$$

Now we just need to choose V_1 so that F is a diffeomorphism onto its image, for which we can use the implicit function theorem. It suffices to show that $\mathrm{d}F$ is bijective at 0. And we can compute, using the definition of geodesics, that at $(0, 0)$,

$$\mathrm{d}F = \begin{pmatrix} \mathrm{d}\varphi & \mathrm{d}\varphi \\ 0 & \mathrm{d}\varphi \end{pmatrix}$$

which is invertible. □

The shortest path between any two points on a closed manifold is always a geodesic, though there are many geodesics of different lengths.

Proposition (Normal Neighborhood fact): Let U_x be a normal neighborhood containing point x with corresponding value ε_x, and let $p, q \in U_x$. Then

(i) There is a unique geodesic passing through p, q with length less than ε_x.

(ii) This geodesic uniquely minimizes distance among all C^1 (and piecewise C^1) curves from p to q.

Proof. (i): Let $\exp(tw)$ and $\exp(tv)$ be two different geodesics going to q. The lengths of these geodesics are $|w|$ and $|v|$, as

$$\int_0^1 |\partial_t \exp(tw)| \, \mathrm{d}t = \int_0^1 |w| = |w|$$

because geodesics have constant velocity. If both $|v|$ and $|w|$ are less than ε_x, then exp is not injective, contradicting the normal neighborhood. And if neither are, then exp is not surjective.

(ii): Suppose we have a curve $\sigma : I \to M$. We can pull this back through exp. Then we use Gauss Lemma *revisit and complete this computation!!!!* □

Of course, if M is compact, then we can find a single ε for which this holds globally. In this case, we can reasonably define a metric on M by saying that $d(p, q)$ is the shortest length of geodesics connecting p and q. One can check using the previous proposition that d satisfies the triangle inequality and is therefore a metric. The unit ball at x in this metric is $\exp_x(B_1(0))$.

Question (Poincaré): For which closed Riemannian manifolds do closed[1] geodesics exist?

Theorem (Hadamard): If $\pi_1(M) \neq 0$, then there is a closed geodesic. WOW!!!!

Proof. Start with a non-contractible loop γ in some homotopy class $[\gamma]$. Among $[\gamma]$ there is a loop with least length (this is nontrivial, but you can get it from a limit of curves with length going to the minimum. This is the "direct method" in Calculus of Variations. Actually, because of the normal neighborhood theorem, we can reduce the search space to piecewise geodesics, which makes things easier). This is a geodesic because any perturbation makes γ longer. □

<u>Geodesic Results:</u>

- (Birkhoff): Any (S^2, g) has a closed geodesic.
- (Lusternic-Fet): Any M has a closed geodesic.
- (Morse): "Most" manifolds have infinitely-many geodesics connecting two distinct points.
- (Senei): All manifolds have infinitely-many geodesics connecting two distinct points.
- (Gromov-Meyer): Most manifolds have infinitely-many closed geodesics.
- (Frances-Bengert,Hingston): Any (S^2, g) has infinitely-many closed geodesics.
- *OPEN*: Do all manifolds have infinitely-many closed geodesics?

2. Curvature

An r-tensor is a map $T : \mathfrak{X}(M)^r \to C^\infty(M)$ which is $C^\infty(M)$-linear. This implies that T is defined locally, i.e. $T(X_1, \ldots, X_r)(p)$ only depends on $X_1(p), \ldots, X_r(p)$.

One example of a 4-tensor is the Riemannian Curvature tensor. Let

$$R(X, Y)(Z) := \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]} Z$$

then the Riemann curvature tensor $\mathcal{R}$ is defined

$$\mathcal{R}(X, Y, Z, W) := (R(X, Y)(Z), W)$$

There are a few identities about $\mathcal{R}$:

<u>Curvature Identities:</u>

- $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$
- $\mathcal{R}(X, Y, Z, W) = -\mathcal{R}(Y, X, Z, W)$
- $\mathcal{R}(X, Y, Z, W) = -\mathcal{R}(X, Y, W, Z)$
- $\mathcal{R}(X, Y, Z, W) = \mathcal{R}(Z, W, X, Y)$